ROBOTICS

COURSE WORK PROPOSAL REPORT

Air Quality Monitoring System

NATIONAL INSTITUTE OF BUSINESS MANGEMENT
HND in Software Engineering

COHNDSE261F

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Group No 6

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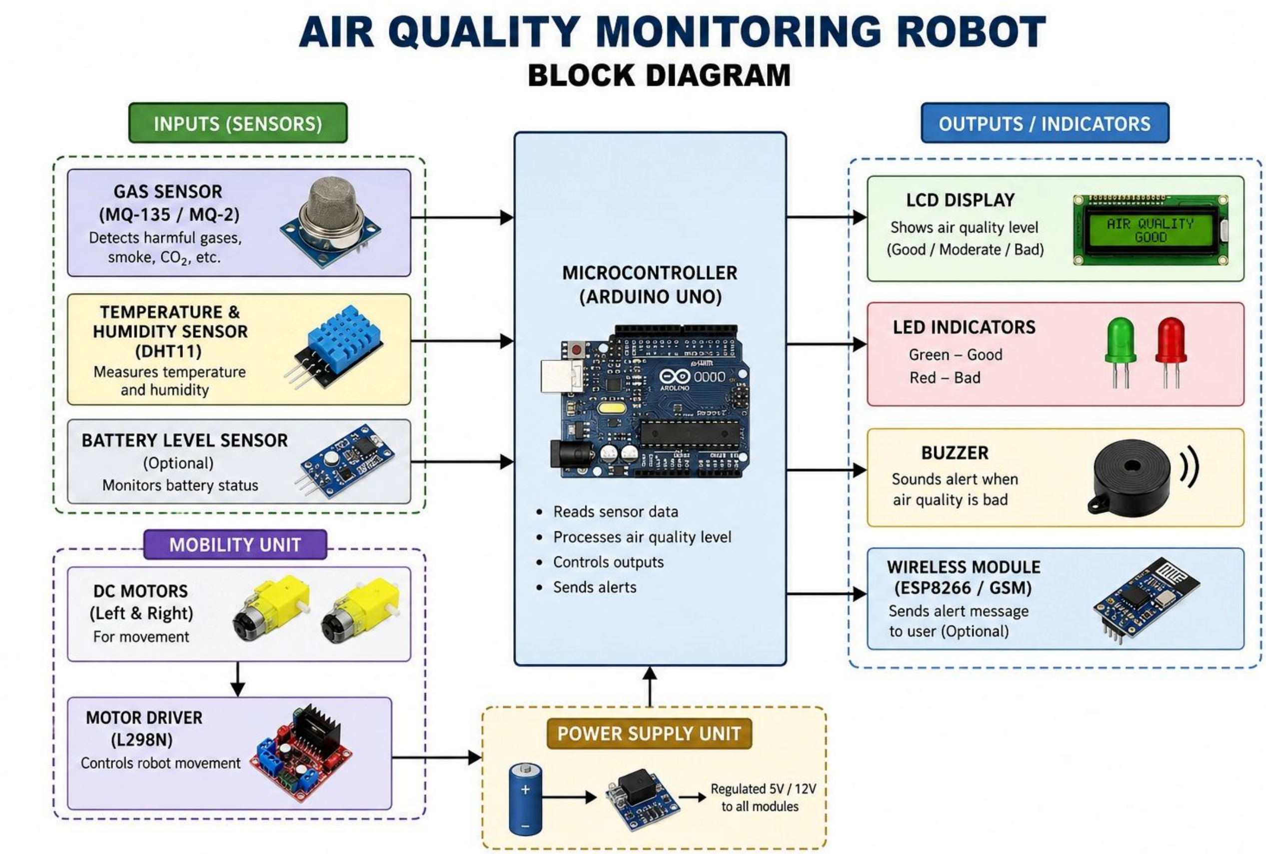
1. Introduction

Air Quality Monitoring Robot is an intelligent system designed to check air pollution levels using gas sensors. It helps to check harmful gases like smoke, carbon dioxide and other toxic gases in the environment. When the pollution levels increases above the safe limit, the system alerts through LED indicators, buzzer alarms, or display messages. This project helps to improve safety and environmental awareness in public and indoor areas such as schools, factories, hospitals, and cities. by continuously monitoring air quality, the robot helps lower health risks caused by air pollution and supports to keep a cleaner and safer environment.

2. Objectives

- To detect air pollution levels using gas sensors.
- To alert users when the air quality becomes dangerous.
- To display current air quality status through indicators or display messages.
- To develop a low-cost and efficient air quality monitoring system.

3. Block Diagram / System Sketch



4. Components & Budget

No	Component	Description	Estimated Cost (LKR)	Image
1	**Main Controller**	ESP32 (Recommended)	**Rs. 2,500 – 4,500**	
2	**Air Quality Sensor**	MQ-135 Sensor	**Rs. 800 – 1,500**	
3	**Optional Sensor**	MQ-2 Smoke/Gas Sensor	**Rs. 600 – 1,200**	
4	**Display System**	LCD 16x2 / OLED Display	**Rs. 800 – 2,500**	
5	**Alert System**	Buzzer + LED Set	**Rs. 200 – 600**	
6	**Power Supply**	Adapter / Power Bank	**Rs. 1,000 – 3,000**	
7	**Cables & Regulator**	USB Cable + Regulator	**Rs. 500 – 1,000**	
8	**IoT Features**	WiFi (ESP32 built-in) + Blynk App	**FREE**	Blynk
9	**Breadboard**	Circuit Testing Board	**Rs. 500 – 1,000**	
10	**Jumper Wires**	Connection Wires	**Rs. 500 – 1,500**	
11	**Casing / Box**	Protective Cover	**Rs. 1,000 – 2,000**	

TOTAL ESTIMATED COST

BASIC VERSION
(No App)
Rs. 5,000 – 10,000

STANDARD STUDENT PROJECT (RECOMMENDED)
Rs. 10,000 – 18,000

SMART IOT VERSION
(With App + WiFi)
Rs. 15,000 – 25,000

5. Gantt Chart